

Predictive Model Plan – Student Template

Use this template to structure your submission. You can copy and paste content from GenAI tools and build around it with your own analysis.

1. Model Logic (Generated with GenAI)

Use a GenAI tool (e.g., ChatGPT, Gemini) to generate the logic or structure of your predictive model.

- You may include pseudo-code, a step-by-step process, or a simplified code snippet.
- Briefly explain what the model is designed to do.

Paste your GenAI-generated output below or describe the logic in your own words:

Model: Logistic Regression

Purpose:

The model predicts whether a customer is likely to become delinquent (miss future payments). It uses historical customer information to calculate a probability score and classify customers as Low Risk or High Risk.

Workflow:

1. Load the dataset.
2. Handle missing values using median/mode imputation.
3. Select important features.
4. Split the data into training and testing sets.
5. Train a Logistic Regression model.
6. Generate a delinquency probability.
7. Classify customers based on a threshold (e.g., 0.5).
8. Evaluate model performance using Accuracy, Precision, Recall, F1 Score, and AUC.

Top 5 Input Features:

- Income
- Credit Utilization
- Number of Missed Payments
- Credit History Length
- Outstanding Loan Balance

2. Justification for Model Choice

Explain why you selected this specific model type (e.g., logistic regression, decision tree, neural network). Consider:

- Accuracy
- Transparency
- Ease of use or implementation
- Relevance for financial prediction
- Suitability for Geldium's business needs

Logistic Regression is appropriate because it is simple, interpretable, and widely used in financial risk prediction. Financial institutions require transparent models that can explain why a customer is classified as high risk. Although more complex models such as Random Forest or XGBoost may achieve slightly higher accuracy, Logistic Regression offers an excellent balance of predictive performance, explainability, ease of deployment, and regulatory compliance. Therefore, it is a suitable choice for Geldium's delinquency prediction use case.

3. Evaluation Strategy

Outline how you would evaluate your model's performance. Include:

- Which metrics you would use (e.g., accuracy, precision, recall, F1 score, AUC)
- How you would interpret those metrics
- Any plans to detect or reduce bias in your model
- Ethical considerations in making predictions about customer financial behavior

Evaluation Metrics:

- Accuracy
- Precision
- Recall
- F1 Score
- AUC-ROC

Interpretation:

- Accuracy measures overall correctness.
- Precision reduces false positives.
- Recall ensures most high-risk customers are identified.
- F1 Score balances precision and recall.
- AUC-ROC measures how well the model separates risky and non-risky customers.

Bias & Fairness:

- Compare results across customer groups.
- Monitor demographic parity and disparate impact.
- Retrain the model regularly with updated data.

Ethical Considerations:

- Protect customer privacy.
- Avoid discriminatory outcomes.
- Keep predictions transparent and allow human review for important financial decisions.